Continual Object Detection in Emerging Modalities - Events Dataset

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Abstract. Event cameras produce asynchronous, high-temporal-resolution visual streams with sub-millisecond latency, high dynamic range and strong robustness to motion blur, making them attractive for time-critical perception tasks such as autonomous driving. However, they suffer from a catastrophic performance drop on the originally trained classes when incrementally learning new classes without revisiting the old data. In this paper we study class-incremental continual learning for event-based object detection by combining the Recurrent Vision Transformer (RVT) backbone with mechanisms inspired by Static-Dynamic Co-Teaching (SDCoT). Concretely, we adapt SDCoT's static-teacher pseudo-labeling and distillation concepts to RVT and evaluate four incremental protocols (fine-tuning, freeze&add, supervised mixed-label training, and supervised+distillation) on the 1 Megapixel (1 Mpx) event dataset. Experiments are reported in mAP@50 on the official test split; inference latency remains comparable to the original RVT ($<$ 12 ms on a T4 GPU). The findings demonstrate that SDCoT-style static guidance is effective for event-based detectors and motivates future work for further gains.

Keywords: Event cameras, Continual learning, Class-incremental learning, Recurrent Vision Transformer, SDCoT

1 Introduction

Event cameras have recently emerged as powerful alternatives to conventional frame-based sensors by offering sub-millisecond latency, high dynamic range, and strong robustness to motion blur. These unique properties make them particularly well-suited for safety-critical applications such as autonomous driving, where rapid perception and decision-making are essential. Recurrent Vision Transformers (RVTs) [5] have demonstrated state-of-the-art performance in object detection with event data by leveraging convolutional priors, attention-based spatial feature interaction, and recurrent temporal aggregation to achieve both efficiency and accuracy. Despite these advances, current RVT-based approaches are trained in a static setting, assuming a fixed set of object categories. In real-world deployment, however, perception systems must continually adapt to new classes without catastrophic forgetting of previously learned ones.

Catastrophic forgetting, the tendency of deep neural networks to severely degrade in performance on old classes when trained on new ones, poses a significant challenge for continual learning. This issue has been extensively studied in image classification and, more recently, in 3D object detection. Static-Dynamic Co-Teaching (SDCoT) [13] was introduced as the first framework to address class-incremental learning in 3D object detection . By incorporating a static teacher to preserve old knowledge through pseudo-labels and distillation loss, and a dynamic teacher to exploit new data through consistency regularization, SDCoT strikes a balance between stability and plasticity. However, the application of such continual learning strategies to event-based object detection remains unexplored.

In this work, we bridge this gap by adapting the SDCoT framework to RVTs for class-incremental object detection with event cameras. Specifically, we enable RVTs to incrementally learn novel object categories while retaining detection performance on base classes. Our framework integrates the static-dynamic teacher-student paradigm with RVT's hierarchical recurrent backbone, thereby addressing the challenge of catastrophic forgetting. Through the combination of supervised and distillation losses, our method enables RVT to adapt to new categories in a class-incremental setting without revisiting old data.

1.1 Contributions

Our main contributions are:

- We present the first study of class-incremental continual learning for event-based object detection, extending Recurrent Vision Transformers (RVTs) [5] with mechanisms inspired by Self-Distilled Continual Transformer (SDCoT) [13].
- We evaluate our approach on an event-based automotive detection scenario, demonstrating that our method maintains strong performance on base classes while effectively adapting to novel categories.

2 Related Work

2.1 Recurrent Vision Transformer (RVT)

The Recurrent Vision Transformer (RVT) [5] is a model designed for event-based object detection, where input comes from event cameras that asynchronously record brightness changes instead of conventional image frames. Each event is defined by pixel coordinates, timestamp, and a polarity. The polarity can be 0 or 1 depending on the sign of the brightness change. Modern event cameras can produce 10s of millions of events per second which renders event-by-event processing out of reach on conventional processing units.

To process the stream of events produced by an event camera, RVT [5] employs a preprocessing step that transforms the raw events into a tensor representation compatible with convolutional networks. Each event is defined as (x_k, y_k, t_k, p_k), where (x_k, y_k) are the pixel coordinates, t_k is the timestamp, and $p_k \in \{0, 1\}$ denotes the polarity of the brightness change. Since events arrive as a continuous sequence rather than in dense frames, they are discretized over time into a fixed window $[t_a, t_b)$.

First, the interval $[t_a, t_b)$ is divided into T equal bins, and each event is assigned to one of these bins according to its normalized timestamp

$$\tau_k = \frac{t_k - t_a}{t_b - t_a} \cdot T.$$

For each bin, two separate channels are maintained: one for positive polarity events and one for negative polarity events. This yields a four-dimensional tensor

$$E(p, \tau, x, y) = \sum_{e_k \in E} \delta(p - p_k)\, \delta(x - x_k, y - y_k)\, \delta(\tau - \tau_k),$$

where p denotes polarity, τ the temporal bin, and (x, y) the spatial location. In words, $E(p, \tau, x, y)$ stores the count of positive or negative events occurring at pixel (x, y) within the temporal bin τ.

As a final step, the polarity and temporal dimensions are flattened, resulting in a three-dimensional tensor of shape $(2T, H, W)$. This representation corresponds to a stack of T two-channel frames and is directly compatible with standard 2D convolutions, thereby allowing RVT to leverage efficient convolutional backbones while preserving the spatio-temporal structure of the original event stream.

The RVT backbone is organized hierarchically into four stages, each composed of RVT blocks. An RVT block contains three main components:

1. **Convolutional prior:** a lightweight convolution with overlapping kernels downsamples spatial resolution and simultaneously injects positional priors, acting as conditional positional embeddings [1].
2. **Dual attention mechanisms:** each block applies local self-attention in small non-overlapping windows to capture fine-grained details [12], followed by dilated global attention across grid partitions to exchange information over large spatial extents with linear complexity.
3. **Temporal recurrence with LSTMs:** plain LSTM cells are used at the end of each stage to aggregate temporal information across event slices [8]. The hidden and cell states (h, c) are propagated across time steps, enabling recurrent modeling of temporal context. Unlike heavier Conv-LSTM layers, RVT's feature-wise LSTMs (equivalent to 1×1 convolutions) reduce computation and parameters while maintaining long-range temporal dependencies.

The four stages progressively downsample the feature maps (with strides of 4, 8, 16, and 32) while increasing channel depth. Hidden states from stages two through four are reshaped into 2D feature maps and provided to the detection framework.

For detection, RVT integrates with the YOLOX [4] one-stage detector. The YOLOX head applies a standard combination of IoU loss for bounding box regression, classification loss, and localization loss, averaged across spatial batch and temporal sequence dimensions. In this setup, RVT supplies temporally coherent features for detection at each event slice.

Overall, RVT achieves strong results: it reaches 47.2% mAP on the Gen1 automotive dataset [3] and 47.4% mAP on the 1 Mpx dataset [7], while running at less than 12 ms per inference on a T4 GPU. Its efficient attention design and lightweight LSTM recurrence make it both computationally efficient and parameter-efficient (e.g. 18.5M parameters for the base model), enabling real-time event-based object detection.

2.2 Static-Dynamic Co-Teaching (SDCoT)

Class-incremental learning (CIL) addresses the practical setting where a detector is first trained on a set of *base* classes C_{base} and later must incorporate *novel* classes C_{novel} that become available sequentially. The central challenge in CIL is *catastrophic forgetting*: when a model is updated on novel classes using only the new data, it typically loses performance on previously learned base classes because those classes are no longer supervised during the incremental phase. For object detection this issue is aggravated by the fact that new training samples often contain unlabeled instances of base classes; a naive incremental learner will treat such base-class instances as background and thereby accelerate forgetting.

Static-Dynamic Co-Teaching (SDCoT) was proposed to solve class-incremental 3D object detection on point clouds [13]. SDCoT augments a standard 3D detector, VoteNet backbone [9], with a teacher-student co-teaching framework consisting of three networks: the *student* (the model being trained incrementally), a *static teacher* (a frozen copy of the base model), and a *dynamic teacher* (an exponential moving average (EMA) of the student). The static teacher provides stable, historical knowledge about base classes and generates pseudo-labels for base-class instances present in novel data; the dynamic teacher provides adaptive guidance by enforcing prediction consistency across perturbations and over the course of training.

To avoid treating base-class instances inside novel training samples as background, SDCoT uses the static teacher to generate pseudo annotations for base classes. Given an input sample from D_{novel} (the dataset containing novel-class labels only), the static teacher predicts a set of proposals and these are filtered using two thresholds: an objectness threshold τ_o and a classification confidence threshold τ_c. Selected proposals become pseudo labels $\hat{Y}_{\text{base}}$ for base classes. Because VoteNet [9] uses stochastic subsampling (seed and vote sampling) the proposal sets produced by different networks can be unaligned; SDCoT solves this by reusing the sampled seed/vote indices from the student when running the static teacher so that proposals can be compared and used consistently across networks.

SDCoT trains the student with three complementary objectives:

1. **Supervised loss** $\mathcal{L}_{\text{sup}}$: standard detection loss (classification + box regression) computed with the *mixed labels* consisting of pseudo labels for base classes and ground-truth labels for novel classes.
2. **Distillation loss** $\mathcal{L}_{\text{dis}}$: a feature/prediction distillation term that forces the student's logits for base classes to remain close to the static teacher's logits. SDCoT normalizes class logits by subtracting the mean over the class dimension and computes squared L_2 distance across K proposals:

$$\mathcal{L}_{\text{dis}} = \frac{1}{K}\sum_{i=1}^{K} \left\| \bar{p}^{S}_{B,i} - \bar{p}^{T}_{B,i} \right\|_2^2,$$

   where $\bar{p}^{S}_{B,i}$ and $\bar{p}^{T}_{B,i}$ are the normalized classification logits for the i-th proposal on base classes produced by the student and static teacher, respectively.
3. **Consistency loss** $\mathcal{L}_{\text{con}}$: a teacher-student consistency regularizer computed between the student and the dynamic teacher (EMA of the student). The dynamic teacher is obtained by exponential averaging of student weights:

$$\Phi'_t = \alpha\, \Phi'_{t-1} + (1-\alpha)\, \Phi_t,$$

   where α is the EMA decay. The consistency loss encourages the student to produce invariant predictions under augmentations/perturbations and helps exploit the underlying structure of novel data.

The three losses are combined as a weighted sum:

$$\mathcal{L} = \lambda_s \mathcal{L}_{\text{sup}} + \lambda_d \mathcal{L}_{\text{dis}} + \lambda_c \mathcal{L}_{\text{con}}.$$

VoteNet's [9] original prediction head includes a fixed-size set of outputs tied to the number of classes. SDCoT adapts VoteNet to the incremental setting by decoupling the final MLP into a regressor branch and a classifier branch, and by dynamically adding new classifier weights for novel classes when they arrive. Furthermore, SDCoT replaces class-aware box size predictions with a class-agnostic size prediction to simplify adding novel categories. These changes make the detector amenable to growing the label set without retraining from scratch.

SDCoT is evaluated on benchmark 3D detection datasets (SUN RGB-D [11] and ScanNet [2]) under multiple incremental settings (batch and sequential additions of novel classes). Empirically, SDCoT substantially reduces forgetting compared to naive baselines such as freeze & add and plain fine-tuning.

The SDCoT framework thus provides a practical, effective recipe for class-incremental 3D object detection by combining pseudo-labelling, deep distillation, and consistency regularization in a teacher-student co-teaching paradigm built around a VoteNet detector [9].

3 Approach

In this work we adapt the Static-Dynamic Co-Teaching (SDCoT) class-incremental learning paradigm to an event-based detector by applying it to the Recurrent Vision Transformer (RVT) [5] backbone. The motivation is to enable an event-based detector to learn novel object categories incrementally while preserving performance on previously learned base classes. Specifically, we adopt the static-teacher idea and the pseudo-labeling plus distillation mechanism from SDCoT [13]. The static teacher in our experiments is the pretrained RVT checkpoint provided with the dataset; it is used to produce pseudo annotations for base-class instances that co-occur in the novel training samples. We evaluated four experimental conditions:

1. **Fine-tuning.** Initialize the model from the pretrained RVT checkpoint, append a new classifier head for novel classes and fine-tune all weights (excluding classification head of base classes) on the novel-class training data (no pseudo labels or distillation from the static teacher).
2. **Freeze & Add.** Freeze the pretrained RVT model, append a new classifier head for novel classes and train only the newly added parameters on the novel data.
3. **Supervised loss only (CIL with $\mathcal{L}_{\text{sup}}$).** Initialize the student from the pretrained checkpoint and train on the novel data using supervised detection loss computed on *mixed labels* (i.e., ground-truth novel labels plus pseudo labels for base classes generated by the static teacher).
4. **Supervised + distillation losses (CIL with $\mathcal{L}_{\text{sup}} + \mathcal{L}_{\text{dis}}$).** Same as (3) but additionally apply a distillation loss that encourages the student's base-class logits to remain close to the static teacher's logits.

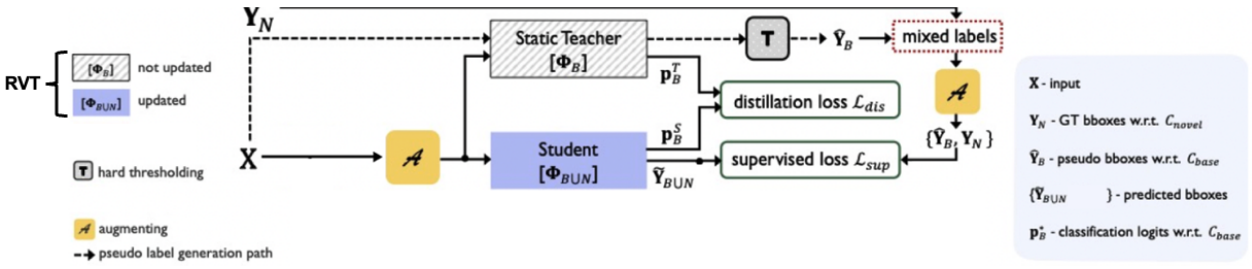


Fig. 1: SDCoT architecture with static teacher and student model. The student and teacher network are RVT detectors.

As far as the RVT models are concered in our experiments, we follow the predefined RVT preprocessing and training protocol [5]. Incoming events are aggregated into 50 ms temporal windows and discretized into

$T = 10$ bins, producing input tensors of shape $(2T, H, W)$ per slice (two polarity channels per bin). For experiments that use mixed supervision (3) and (4), we run the frozen pretrained RVT (static teacher) on each novel training sample to produce base-class proposals. These pseudo labels are combined with the ground-truth annotations for novel classes to form the mixed supervision used by the supervised loss $\mathcal{L}_{\text{sup}}$. The supervised detection loss $\mathcal{L}_{\text{sup}}$ is the same multi-task YOLOX detection loss [4] used in the RVT pipeline: a combination of IoU (or localization) loss for bounding boxes, classification loss for categories, and regression loss, averaged over both spatial batch and sequence length. To accommodate the larger label set we extend the YOLOX [4] detection head so that its classification output dimension equals the total number of classes considered during evaluation (base + novel). Concretely, the classification output layer is expanded and new weights corresponding to novel classes are randomly initialized. This modification enables the student to output scores for both base and novel classes while using the same backbone.

For the distillation term we adopt a formulation analogous to SDCoT [13]: let the student produce classification logits for the proposals and let the static teacher produce logits for base classes. We first normalize logits by subtracting the mean over the class dimension and then define the distillation loss as

$$\mathcal{L}_{\text{dis}} \;=\; \frac{1}{K}\sum_{i=1}^{K}\left\|\bar{p}^{S}_{B,i} - \bar{p}^{T}_{B,i}\right\|_2^2,$$

where $\bar{p}^{S}_{B,i}$ and $\bar{p}^{T}_{B,i}$ are the normalized base-class logits of the i-th proposal from the student and static teacher, respectively, and K is the number of aligned proposals considered. For experiments that include distillation we minimize the combined objective:

$$\mathcal{L} \;=\; \lambda_s \mathcal{L}_{\text{sup}} \;+\; \lambda_d \mathcal{L}_{\text{dis}}.$$

In our experiments, we set both λ_s and λ_d to 1.

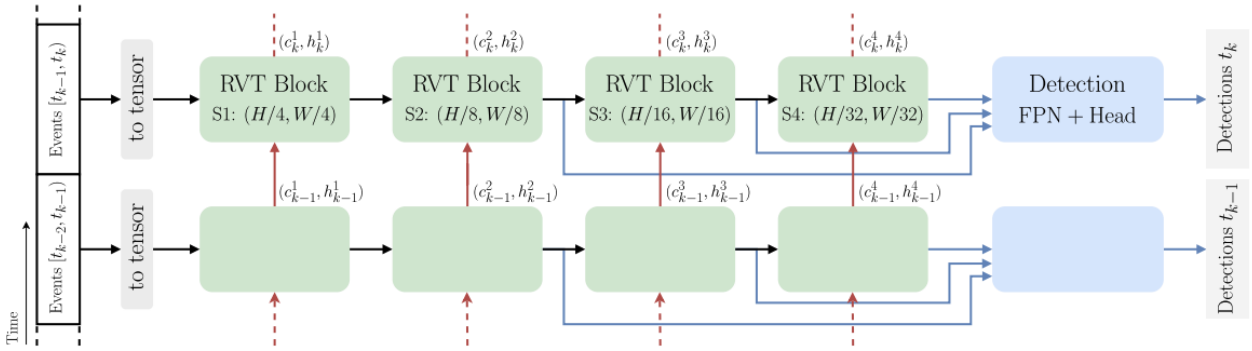


Fig. 2: Events are encoded into tensors and passed through sequential stages, where LSTM states are reused. From the second stage onward, the detection framework uses the LSTM hidden states as features.

4 Experiment & Results

4.1 Dataset

We use the publicly provided 1 Mpx event dataset [7] and the RVT architecture and perform class-incremental experiments. The dataset provides annotations for several traffic classes including *pedestrian*, *two-wheeler*, *car*, *truck*, *bus*, *traffic sign*, and *traffic light*. We use *pedestrian*, *two-wheeler*, *car* to serve as **base** classes and *truck*, *bus* are treated as **novel** classes. A pretrained checkpoint for RVT trained on *pedestrian*, *two-wheeler*, *car* is available; we use this checkpoint both as the initialization for our student models and as the *static teacher* in experiments that require pseudo-labeling. The novel classes for the incremental stage are *truck* and *bus*; during incremental training annotations are available only for these novel classes (i.e., the typical class-incremental setting).

4.2 Training Procedure

Training follows the RVT recipe and augmentations to ensure comparability and reproducibility. Specifically, we use the ADAM optimizer [6] together with a OneCycle learning-rate schedule [10] and mixed-precision training. Sequences of length five are used, with the same augmentations (random horizontal flip, zoom-in and zoom-out). We also adopt the mixed batching strategy from RVT: half of each batch is trained with full backpropagation through time (BPTT) and the other half with truncated BPTT (TBPTT), since it resulted in the best reuslts. We use the RVT default peak learning rate for the 1 Mpx setting (3.5×10^{-4}) and the same total number of iterations. All backbone and classification head weights of base classes are initialized from the provided RVT checkpoint trained on base classes; weights for newly added classifier outputs corresponding to novel classes are randomly initialized. This steps that differences in our setup stem from the incremental training protocol rather than backbone variations. Final predictions at test time are produced by the trained student network.

4.3 Quantitative Results

The experiments are intended to isolate the effect of (i) fine-tuning, (ii) freezing base parameters and only training newly added heads, (iii) incremental supervised learning with pseudo-labels, and (iv) incremental supervised learning augmented with distillation from a static teacher. By reusing the RVT training recipe and the provided pretrained base checkpoint as the static teacher, our setup directly tests the effectiveness of SDCoT-style class incremental learning for event-based detectors and quantifies the extent to which catastrophic forgetting can be mitigated in the case of event based data. We report detection performance using mean average precision at IoU = 0.50 (mAP@50) on the test split of the 1 Mpx event dataset. Since our experiments reuse the RVT backbone and runtime settings from it; consequently the inference latency of our trained models remains effectively the same as reported for the original RVT (< 12 ms on a T4 GPU), we therefore focus this section on accuracy comparisons under the class-incremental setting. Table 1 reports per-class scores for the base and novel categories, while Table 2 summarizes results aggregated over base, novel and all classes.

Table 1: Comparison of class-wise scores on the test set of the 1 Mpx dataset.

	Method	**Pedestrians**	**Two Wheelers**	**Cars**	**Trucks**	**Buses**
1	Base Training	32.0	46.7	63.3	–	–
2	Freeze & Add	32.2	46.7	63.3	0.005	0.0
3	Finetuning	0.0	0.0	0.0	37.8	32.5
4	CIL with $\mathcal{L}_{\text{sup}}$	13.7	21.3	44.9	31.1	26.9
5	CIL with $\mathcal{L}_{\text{dist}} + \mathcal{L}_{\text{sup}}$	11.8	26.1	43.9	32.0	22.6

Table 2: Comparison of base, novel, and overall scores on the test set of the 1 Mpx dataset.

	Method	**Base**	**Novel**	**All**
1	Base Training	47.4	–	–
2	Freeze & Add	47.4	0.0025	28.4
3	Finetuning	0.0	35.2	14.1
4	CIL with $\mathcal{L}_{\text{sup}}$	**26.6**	**29.0**	**27.6**
5	CIL with $\mathcal{L}_{\text{dist}} + \mathcal{L}_{\text{sup}}$	**27.3**	**27.3**	**27.3**

4.4 Qualitative Results

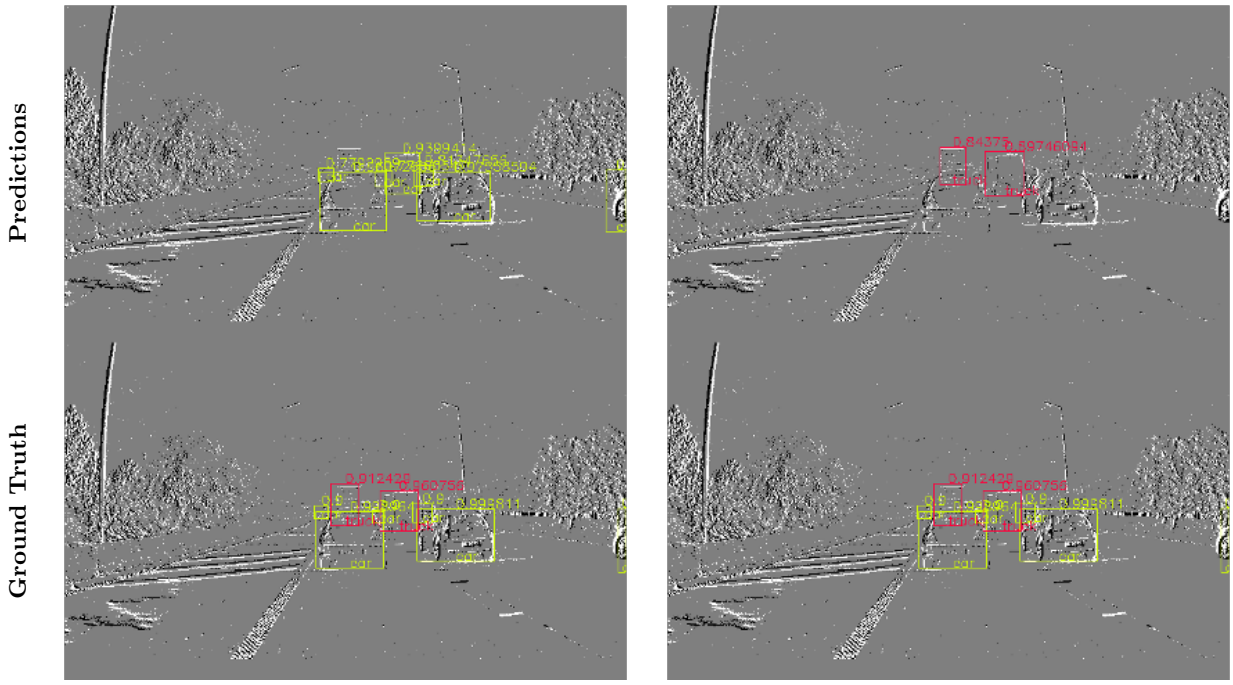


Fig. 3: Visual comparison of predictions (top row) and ground truth (bottom row) for two incremental strategies on the 1 Mpx event dataset. **Left:** Freeze & Add — note that base-class detections (cars, pedestrians, etc.) are preserved but novel classes (truck, bus) are not detected. **Right:** Finetuning — the model fits the novel classes (truck, bus) but suffers from catastrophic forgetting of base classes (base detections disappear).

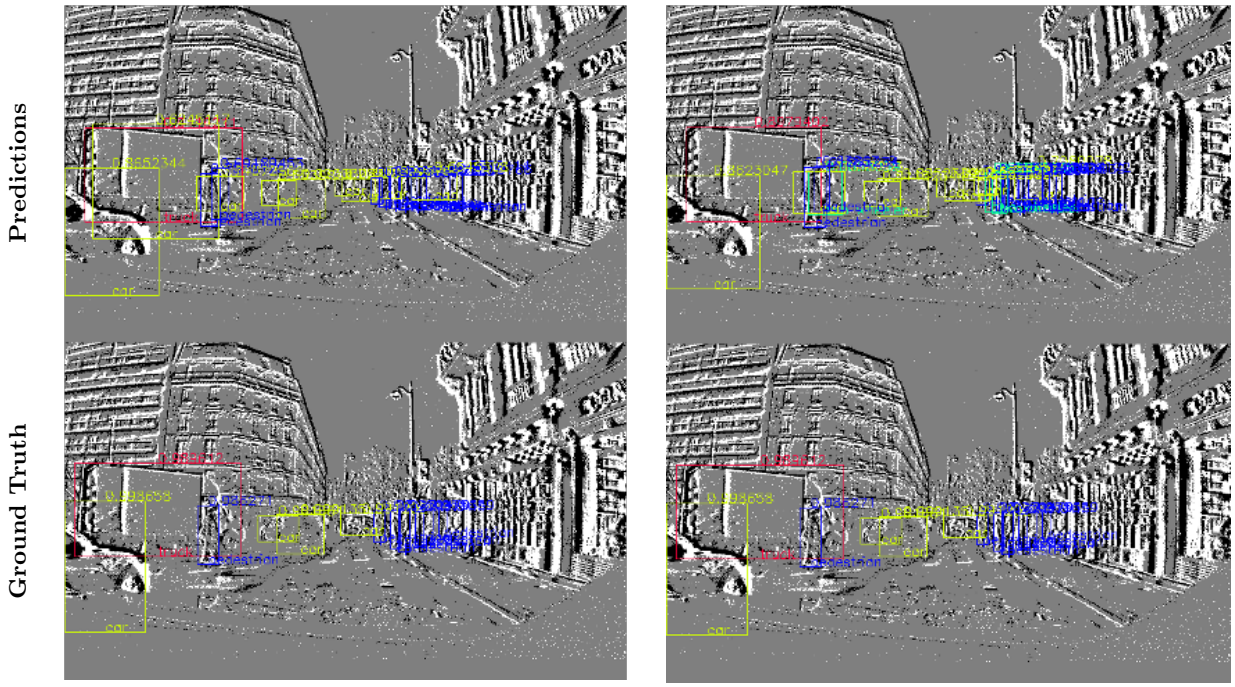


Fig. 4: The models are able to successfully detect both base classes such as cars and pedestrians, as well as novel classes like trucks, demonstrating the effectiveness of the class incremental learning strategy.

5 Conclusion

The results reveal several clear trends. Naive strategies fail in different ways: Freeze & Add preserves base-class performance but is unable to detect novel classes (near-zero novel mAP), while Fine-tuning successfully fits novel classes at the expense of catastrophic forgetting of base classes (base mAP collapses to near zero). Training with mixed supervision (pseudo-labels) using only the supervised detection loss ($\mathcal{L}_{\text{sup}}$) substantially mitigates forgetting and enables learning of novel categories. Adding a distillation term ($\mathcal{L}_{\text{dis}} + \mathcal{L}_{\text{sup}}$), i.e. encouraging the student to match the static teacher's base-class logits, further improves base-class retention (compare base mAP 27.3 vs. 26.6) while maintaining comparable novel-class performance. Note that these experiments exclude the dynamic teacher / consistency loss component from SDCoT; the present findings therefore indicate that static-teacher pseudo-labeling and distillation already provide strong protection against forgetting in event-based RVT detectors, and suggest that incorporating a dynamic teacher is a promising direction for future improvement.

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